Reverse Proxy and TLS

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Reverse Proxy & TLS Overview

Goal

Attendees install and configure Nginx as a reverse proxy and enable TLS (self-signed for testing, or Let's Encrypt for public domains).

Duration

90 minutes

What is a Reverse Proxy?

- Acts as an intermediary for requests from clients seeking resources from servers.
- Protects the underlying service.
- Handles load balancing and TLS termination.
- Allows multiple services to share Port 80/443 mapping on the same machine.

Installing Nginx

- Nginx will run on the host and forward traffic to the local container.

```
sudo dnf install nginx
sudo systemctl enable --now nginx
```

Nginx Configuration

- Create a virtual host configuration `/etc/nginx/conf.d/my-flask-app.conf`:

```
server {
    listen 80;
    server_name your.public.domain; # Use your domain or
↪   IP

    location / {
        proxy_pass http://127.0.0.1:5000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }
}
```

Start Nginx

Enable the site snippet:

```
sudo nginx -t
sudo systemctl restart nginx
```

Transport Layer Security (TLS)

- Ensures data sent to/from the service is encrypted.
- Provides authentication that the website is genuine.
- Needs Certificates (public key/private key pair).

Self-signed vs Let's Encrypt

- Self-Signed Certificates are good for testing (Browsers will show warnings).
- Let's Encrypt provides free, automated, and open certificates using the ACME protocol.

Getting a Let's Encrypt Certificate

Install Certbot and its Nginx plugin:

```
sudo dnf install certbot python3-certbot-nginx
```

Run Certbot to configure Nginx automatically:

```
sudo certbot --nginx -d yourdomain.com # Replace with your
↪  actual domain or IP
```

Final Validation

- Access your service securely:

```
curl https://your.public.domain -X POST -H "Content-Type:
↪  application/json" -d '{"secret_key":
↪  "supersecretkey"}'
```